Intro To Machine Learning • BA4

1 Introduction

ML Basics

2 Linear Supervised ML

Linear regression
Linear models for classification

3 Non linear Supervised ML

k-Nearest Neighbors
Feature expansion
Kernel methods
Artificial neural networks (deep learning)

4 Unsupervised ML

Dimensionality reduction
k-Means Clustering

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Key definiti2ons

Linear map

$T : V \to W$ is linear if $T(\alpha x + \beta y) = \alpha T(x) + \beta T(y)$ for all $x, y \in V$, $\alpha, \beta \in \mathbb{R}$.

Rank–nullity

For $T : V \to W$ with $\dim V < \infty$:

$$\dim \ker T + \dim \operatorname{im} T \;=\; \dim V.$$

Formulas to remember

- **Quadratic:** $x = \dfrac{-b \pm \sqrt{b^2 - 4ac}}{2a}$
- **Euler:** $e^{i\theta} = \cos\theta + i \sin\theta$
- **Geom. sum:** $\sum_{k=0}^{n} r^k = \dfrac{1 - r^{n+1}}{1 - r}$, $r \neq 1$
- **Taylor:** $f(x) = \sum_{k \geq 0} \dfrac{f^{(k)}(a)}{k!}(x - a)^k$

Pitfall

$\log(ab) = \log a + \log b$ *only* when $a, b > 0$.

Useful identities

$$\sin^2\theta + \cos^2\theta = 1$$
$$\cos(2\theta) = \cos^2\theta - \sin^2\theta$$
$$\sin(2\theta) = 2\sin\theta\cos\theta$$

6 Topic 2 — Probability

Basics

- $\mathbb{P}(A \cup B) = \mathbb{P}(A) + \mathbb{P}(B) - \mathbb{P}(A \cap B)$
- $\mathbb{P}(A \mid B) = \dfrac{\mathbb{P}(A \cap B)}{\mathbb{P}(B)}$
- Bayes: $\mathbb{P}(A \mid B) = \dfrac{\mathbb{P}(B \mid A)\mathbb{P}(A)}{\mathbb{P}(B)}$

Common distributions

Dist.	PMF/PDF	$\mathbb{E}[X]$
$\mathrm{Bern}(p)$	$p^x(1-p)^{1-x}$	p
$\mathrm{Bin}(n, p)$	$\binom{n}{x} p^x (1-p)^{n-x}$	np
$\mathrm{Poiss}(\lambda)$	$e^{-\lambda}\lambda^x / x!$	λ
$\mathcal{N}(\mu, \sigma^2)$	$\frac{1}{\sigma\sqrt{2\pi}} e^{-(x-\mu)^2/2\sigma^2}$	μ

Worked

$X \sim \mathcal{N}(0, 1)$. Then $\mathbb{P}(\lvert X \rvert < 1) \approx 0.6827$.

7 Topic 3 — Algorithms

Complexity cheat

Op	Cost
Sort (comparison)	$\Theta(n \log n)$
Hash lookup (avg)	$\Theta(1)$
BFS / DFS	$\Theta(V + E)$
Dijkstra (heap)	$\Theta((V + E) \log V)$

Code snippet

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def bsearch(a, x):
    lo, hi = 0, len(a)-1
    while lo <= hi:
        m = (lo+hi)//2
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8 Quick reference

Greek: $\alpha\,\beta\,\gamma\,\delta\,\varepsilon\,\zeta\,\eta\,\theta\,\iota\,\kappa\,\lambda\,\mu\,\nu\,\xi\,\pi\,\rho\,\sigma\,\tau\,\phi\,\chi\,\psi\,\omega$

Sets: $\mathbb{R}\ \mathbb{N}\ \mathbb{Z}\ \mathbb{Q}\ \mathbb{C}$ **Logic:** $\forall\ \exists\ \Rightarrow\ \Leftrightarrow\ \neg$

Derivatives: $(\sin x)' = \cos x$, $(\cos x)' = -\sin x$, $(\ln x)' = 1/x$, $(e^x)' = e^x$, $(x^n)' = nx^{n-1}$.

Integrals: $\int x^n \, dx = \frac{x^{n+1}}{n+1}$, $\int \frac{1}{x} \, dx = \ln\lvert x\rvert$, $\int e^x \, dx = e^x$.

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